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Cross-Model LLM Code Review: Should you use Claude to review Codex or vice versa?

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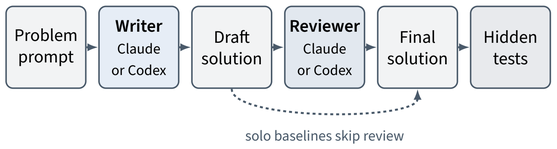
1. Motivation and research questions

Developers increasingly run two coding agents together: one writes a draft, and the other reviews it before the change lands. Leaderboards rank single models in isolation; they do not say which of two available agents should write, which should review, or whether the second pass is worth its cost and time.

We treat this as a role assignment problem between a writer and a reviewer, and measure what a single review pass changes under a common hidden-test evaluator.

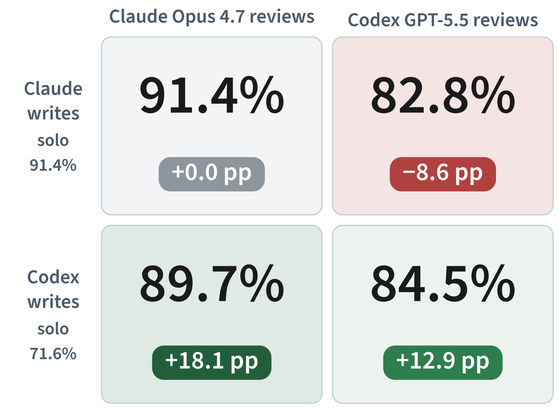
- **RQ1** Does a reviewer pass improve overall correctness?
- **RQ2** Does its value depend on which model writes and which reviews?
- **RQ3** When does the reviewer fix the draft, and when does it regress it?
- **RQ4** Are correctness gains worth the added cost and latency?

2. Study design



- 116 recent hard and medium LiveCodeBench tasks (self-contained Python contest problems scored against hidden tests), all released after both models' training cutoffs; each task has a valid artifact under all six conditions (complete-case pooling).
- Claude Opus 4.7 (Claude Code CLI) and Codex GPT-5.5 (Codex CLI), both at high reasoning effort.
- Static review: the reviewer reads the problem and the writer's draft and must re-emit a complete final program; there is no approve or no-change action. It cannot run code, query a test runner, or inspect execution traces.
- The benchmark scores only the final artifact, so for a fixed writer, the difference between the solo and reviewed conditions is the review pass itself.
- Metrics: pass rate (bootstrap 95% CI), SID, reviewer fixes and regressions; exact McNemar paired tests with Benjamini-Hochberg (BH) correction, defined in the results panels.

3. Pass rate by writer and reviewer



Answer: have Claude Opus 4.7 review Codex GPT-5.5 drafts (+18.1 pp); the reverse loses 8.6 pp. No reviewer improves Claude Opus 4.7 drafts.

4. Complete results

Cond.	Pass rate [95% CI]	SID	Cost	Latency	Reg. rate
Solo baselines					
A	.914 [.862, .957]		$.226	86.2 s	
O	.716 [.629, .793]		$.190	38.5 s	
With a review pass					
AO	.828 [.759, .897]	−.086	$.382	118.0 s	.112
OA	.897 [.836, .948]	+.181	$.443	112.4 s	.043
AA	.914 [.862, .957]	+.000	$.389	135.8 s	.026
OO	.845 [.776, .905]	+.129	$.312	67.9 s	.052

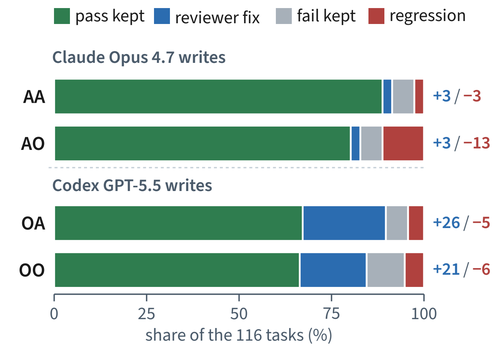
Letters are writer then reviewer (A = Claude Opus 4.7, O = Codex GPT-5.5). SID: change vs the writer's solo baseline. Reg. rate: share where the baseline passed but the reviewed run failed.

5. Six conditions

ID	Writer	Reviewer	Role type
A	Claude Opus 4.7	–	Solo baseline
O	Codex GPT-5.5	–	Solo baseline
AO	Claude Opus 4.7	Codex GPT-5.5	Cross-model
OA	Codex GPT-5.5	Claude Opus 4.7	Cross-model
AA	Claude Opus 4.7	Claude Opus 4.7	Same-model
OO	Codex GPT-5.5	Codex GPT-5.5	Same-model

Two-letter labels are in writer-then-reviewer order: AO is Claude Opus 4.7 output reviewed by Codex GPT-5.5.

6. What the reviewer changed



OA fixes 26 of the 33 failures Codex leaves and regresses 5 (net +21); AO fixes 3 of Claude's 10 and regresses 13 (net −10).

7. Statistical support

Comparison	First only	Second only	p_{BH}	
OA vs O	26	5	.0010	significant
OO vs O	21	6	.0222	significant
AO vs A	3	13	.0456	significant
AA vs A	3	3	1.000	n.s.
AA vs AO	12	2	.0323	significant
AO vs OA	5	13	.1444	n.s.

First/second only: McNemar discordant pairs. AO vs OA does not survive correction; the 19.8 pp baseline gap means reviewer strength and direction cannot be separated.

8. Helpful reviews edit; harmful ones rewrite

Targeted edits that keep the writer's structure

Examples: restoring deleted state in a segment tree while keeping the rest of the writer's interface; removing stray code fences so an otherwise working solution could be parsed.

Rewrites that discard a working draft

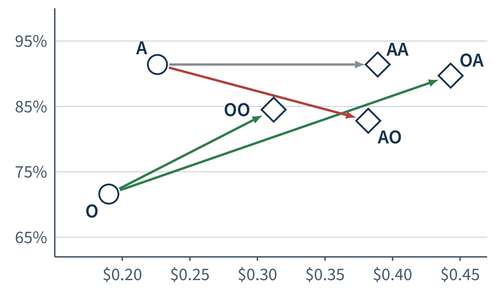
Examples: replacing a passing median-window solution with a heap rewrite that fails hidden tests; removing a range guard that protected an edge case.

Discussion and implications

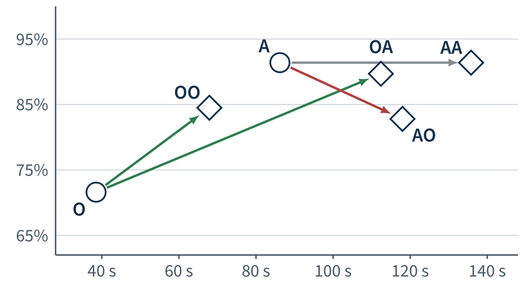
- Review pays where the writer under-verifies: Claude's long solo turns (86.2 s vs 38.5 s) suggest heavy first-pass checking (AA +0.0 pp); Codex leaves catchable errors (OO +12.9 pp).
- Gains track the reviewer's strength relative to the writer, consistent with prior self-repair results; a second pass can subtract as well as add.
- Most AO losses are rewrites of working drafts (an interpretive pattern from inspected cases, not a coded taxonomy). A reviewer prompt that first decides whether to intervene, and keeps the writer's structure by default, is a concrete next step.
- A reviewer able to run even public tests could recover much of what static inspection misses; other model pairs are untested.

9. Cost and latency trade-offs

Pass rate versus cost per task



Pass rate versus latency per task



Circles: solo baselines; diamonds: with reviewer; arrows: the reviewer's effect (green gain, red loss, gray flat); single-run per-task means at current pricing. No reviewed condition beats Claude Opus 4.7 solo on both axes. OA buys each net fixed task for about $1.40; OO about $0.95.

10. Practical guidance

On this benchmark, with one static review pass:

- **Writer must be Codex GPT-5.5:** use Claude Opus 4.7 review (OA) when accuracy dominates (89.7%; the lowest reviewed regression rate, 4.3%). Use self-review (OO) when average per-task latency must stay near 70 s (84.5%).
- **Writer must be Claude Opus 4.7:** skip review. Self-review left the pass rate unchanged in this run (3 fixes, 3 regressions) while adding 72% to cost and 49.6 s of latency; Codex GPT-5.5 review loses 8.6 pp.
- **Both ends free:** Claude Opus 4.7 solo is Pareto-optimal; AA is strictly dominated by A.

Scope and limitations

116 tasks; a single run per condition; one benchmark family; one static review pass with no execution feedback; results are sensitive to prompts and to model and tool versions; reproducing all six conditions costs about $225.

Code and data

Full artifact: prompts, per-task records, and analysis scripts.
github.com/shawnzxiang/cross-model-review-code

A second review pass is not free insurance: its value depends on who wrote the draft and who reviews it.

Code, data, prompts, and per-task records: github.com/shawnzxiang/cross-model-review-code · Supported by US NSF Award #2436427